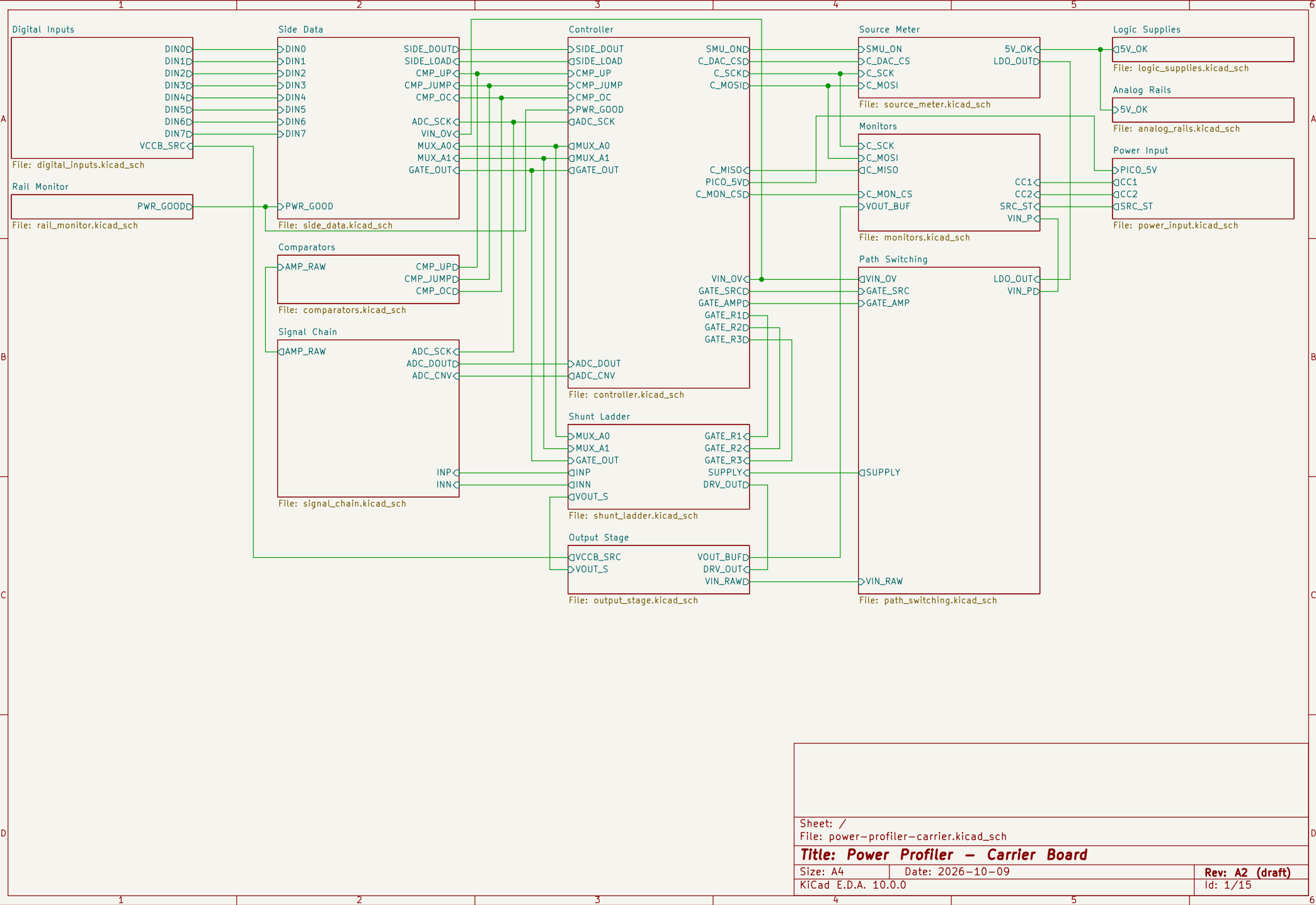
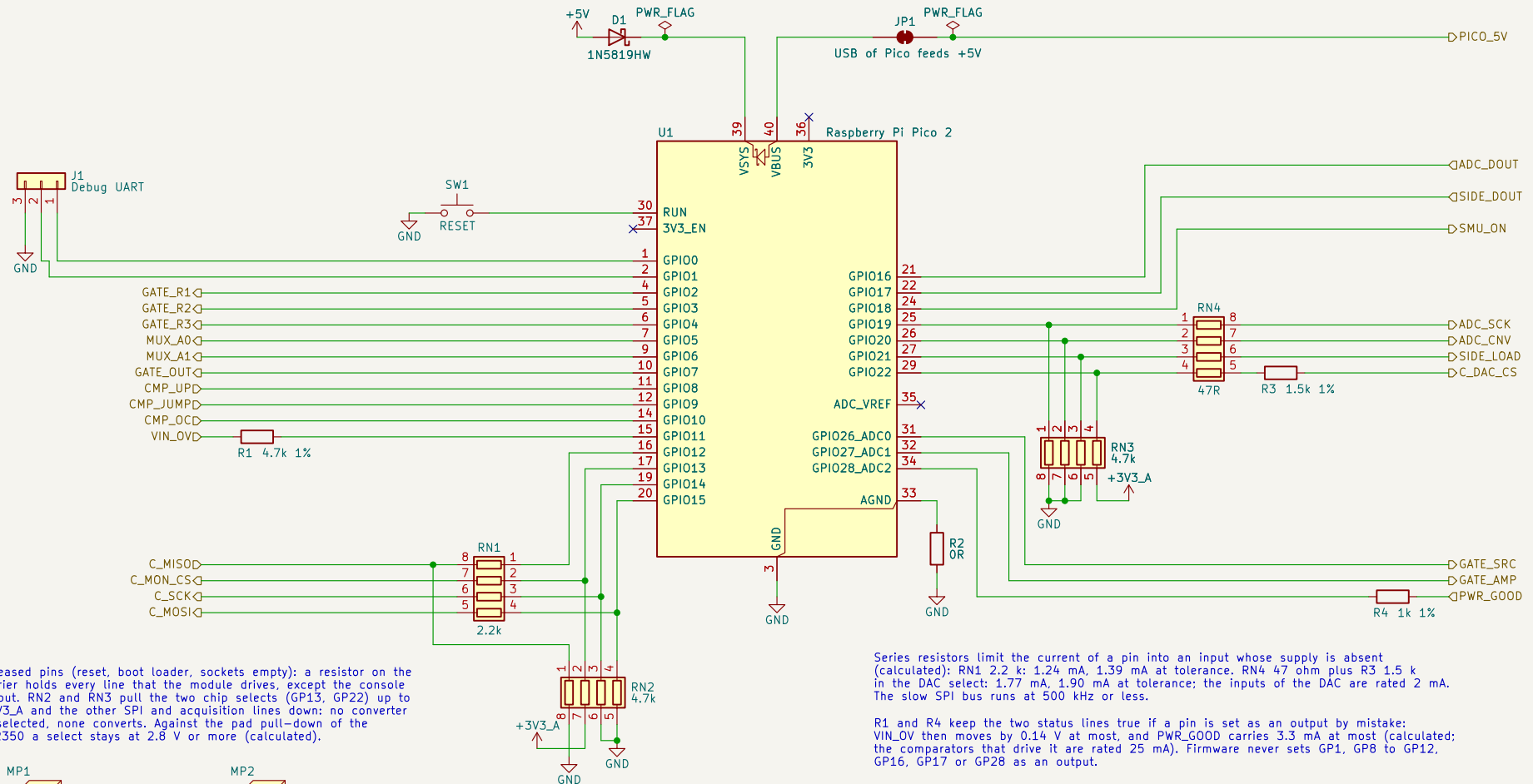




U1 is a Raspberry Pi Pico 2 with headers, plugged into the sockets MP1 and MP2:
the only programmable part. RP2350 stepping A3 or A4 is preferred. A Pico 2 W fits the sockets, but the firmware does not support it.
PIO block 0 (GP16, GP17, GP19 to GP21) clocks the converter and reads its data and the side data in the same word, PIO block 1 (GP2 to GP10) is the range sequencer:
comparators in, gates and multiplexer address out. J1: TX, RX, GND at 3.3 V.

Supply: the USB voltage of the module leaves through JP1 as PICO_5V to its current limiter (0.76 A) and to the source selector on the Power Input sheet, where USB-C has priority. The selected +5V feeds VSYS through D1. Cut JP1 to keep the carrier off the USB port of the computer.



Socket 1x20 2.54 mm



Socket 1x20 2.54 mm

Raspberry Pi Pico 2 on two pin sockets: the only programmable part

Sheet: /Controller/
File: controller.kicad_sch

Title: Controller: Raspberry Pi Pico 2

Size: A4 Date: 2026-10-09

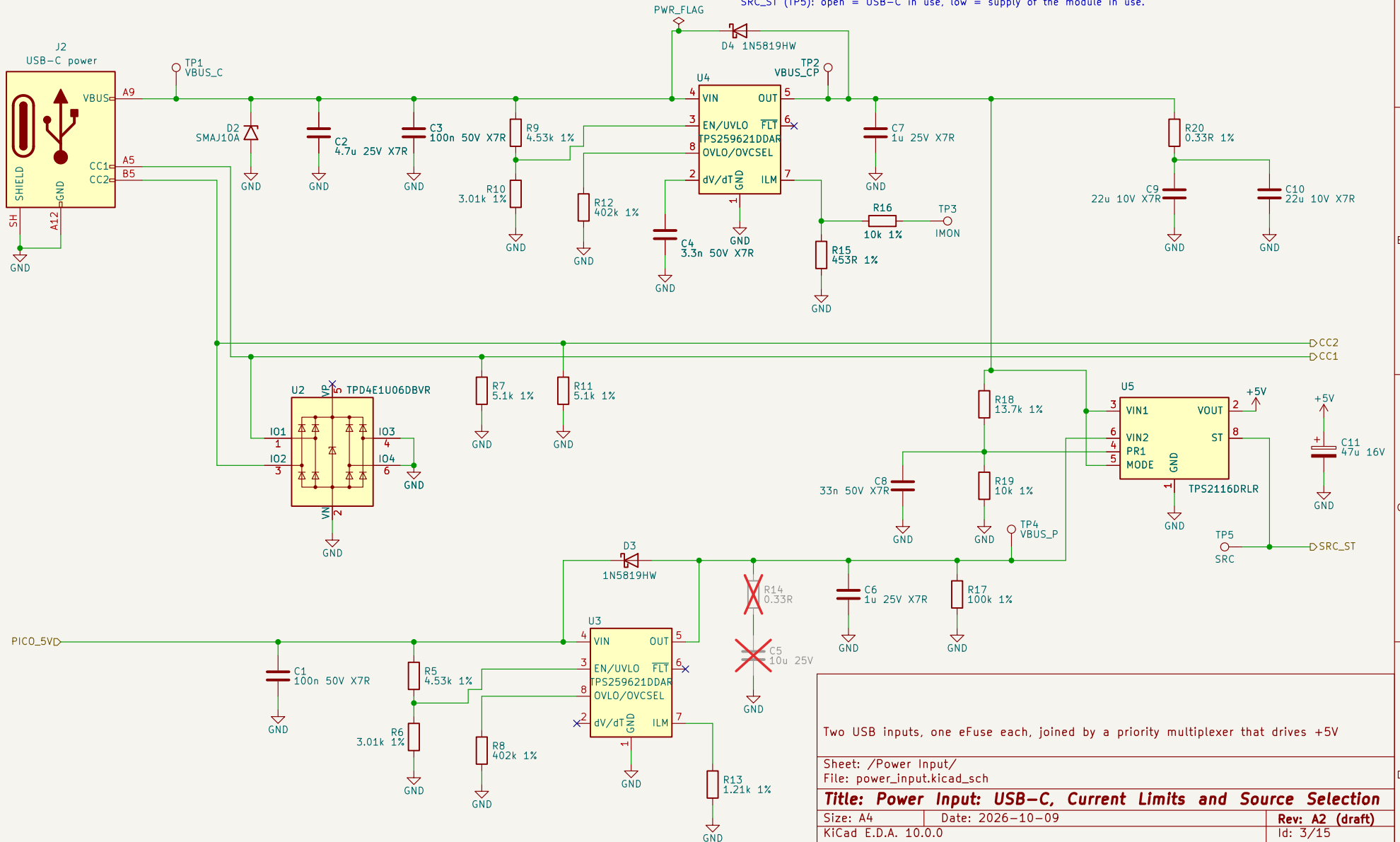
KiCad E.D.A. 10.0.0

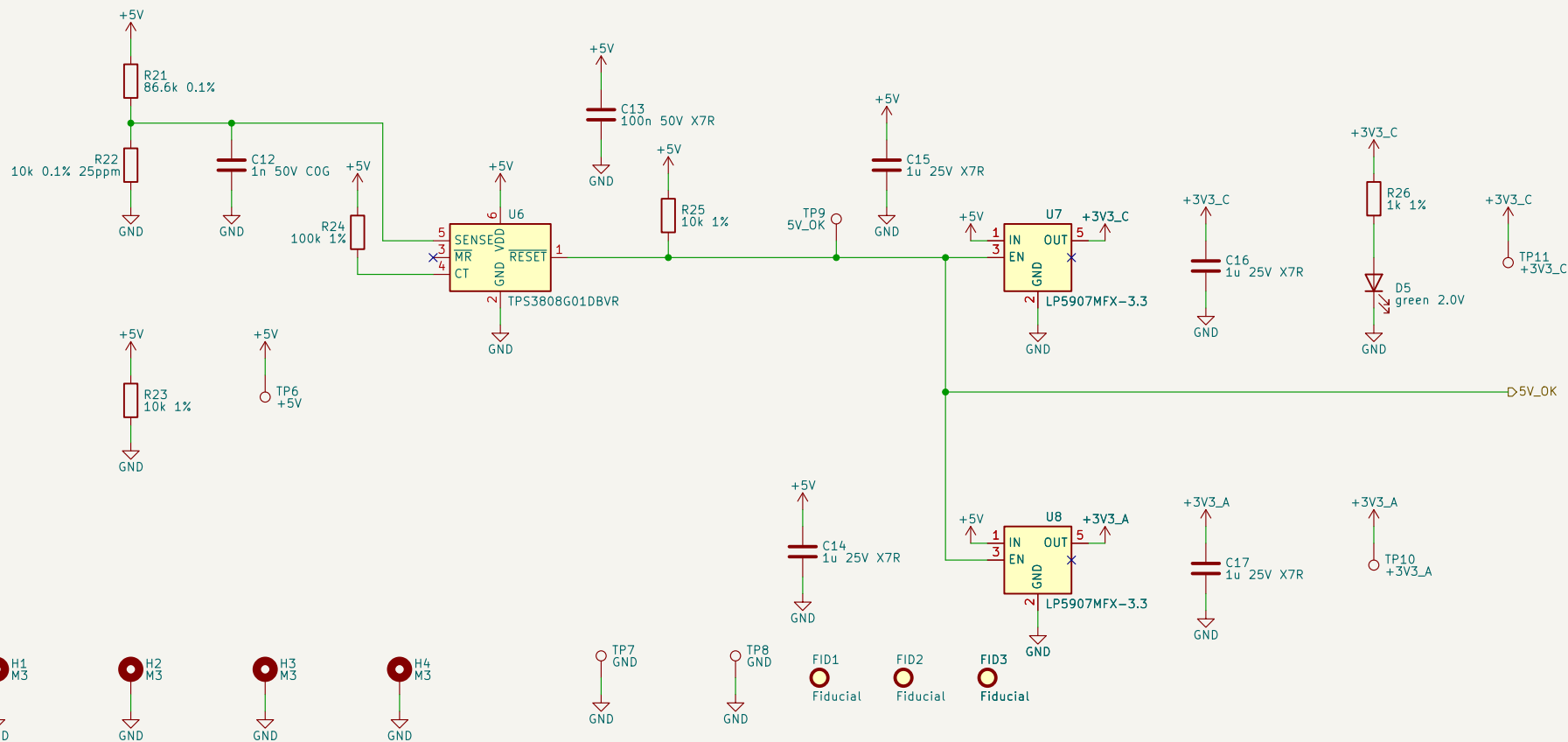
Rev: A2 (draft)

Id: 2/15

J2 is a USB-C sink for 5 V. R7 and R11 pull CC1 and CC2 down; the firmware reads the CC voltage through the monitor converter; below 0.61 V default USB, 0.70 V to 1.16 V 1.5 A, 1.31 V to 2.04 V 3 A. U2 protects the CC pins; D2, C2 and C3 sit at the receptacle, ahead of the current limit. D2 stands off 10 V; the input tolerates no more than that. U4 (eFuse A) turns on when VBUS_C is above 2.92 V to 3.09 V and off below 2.67 V to 2.87 V (R9, R10, which also bleed VBUS_C). It limits the current to 2.0 A (1.81 A to 2.17 A, R15) and clamps its output to 5.28 V to 5.61 V when the input is above 5.54 V to 5.83 V (R12). C4 sets a turn-on delay of about 0.25 ms and a ramp of 13 V/ms (datasheet, typical). TP3 shows the input current, 0.297 V per ampere, behind R16. R20, C9 and C10 damp the ring of a contact that closes; D4 takes their charge back to the input when the plug is pulled, so that the output stays within 0.3 V of the input.

U3 (eFuse B) does the same for the supply from the USB connector of the controller module: same thresholds (R5, R6), limit 0.76 A (0.67 A to 0.85 A, R13), fastest ramp, turn-on delay about 0.08 ms typical. R17 defines VBUS_P while U3 is off. R14, C5: position for a damper of VBUS_P, not fitted. Fit 0.33 ohm and 10 uF if the bench shows VBUS_P above 6.0 V when the data cable is plugged again. U5 drives +5V from USB-C (VBUS_CP) while PR1 is above 1.0 V, that is while VBUS_CP is above 2.37 V (2.15 V to 2.59 V, calculated from R18, R19), otherwise from the module (VBUS_P). It blocks reverse current into both inputs. C8 delays PR1 by 0.19 ms, until the ramp of U4 is nearly complete. Order of the thresholds on a falling rail: supervisor of +5V 3.83 V to 4.00 V (next sheet), eFuse off 2.67 V to 2.87 V, change of source 2.15 V to 2.59 V. The multiplexer leaves USB-C only when U4 is already off. SRC_ST (TP5): open = USB-C in use, low = supply of the module in use.





U6 watches the 5 V rail. 5V_OK goes low when +5V falls below $0.405 \text{ V} \times (1 + R21 / R22) = 3.91 \text{ V}$ (3.83 V to 4.00 V with the tolerances) and returns 0.18 s to 0.42 s after the rail has risen above 4.12 V at most (R24 selects 0.3 s typical). 5V_OK is open drain with R25 to +5V. Low, it switches off U7 and U8 here and, on other sheets, the +12 V regulator, the charge pump and the pre-regulator: the carrier has one off state, in which only +5V and +13V5 stay up. C12 filters SENSE with 9 us. TP9 held to ground keeps the carrier off during bring-up.

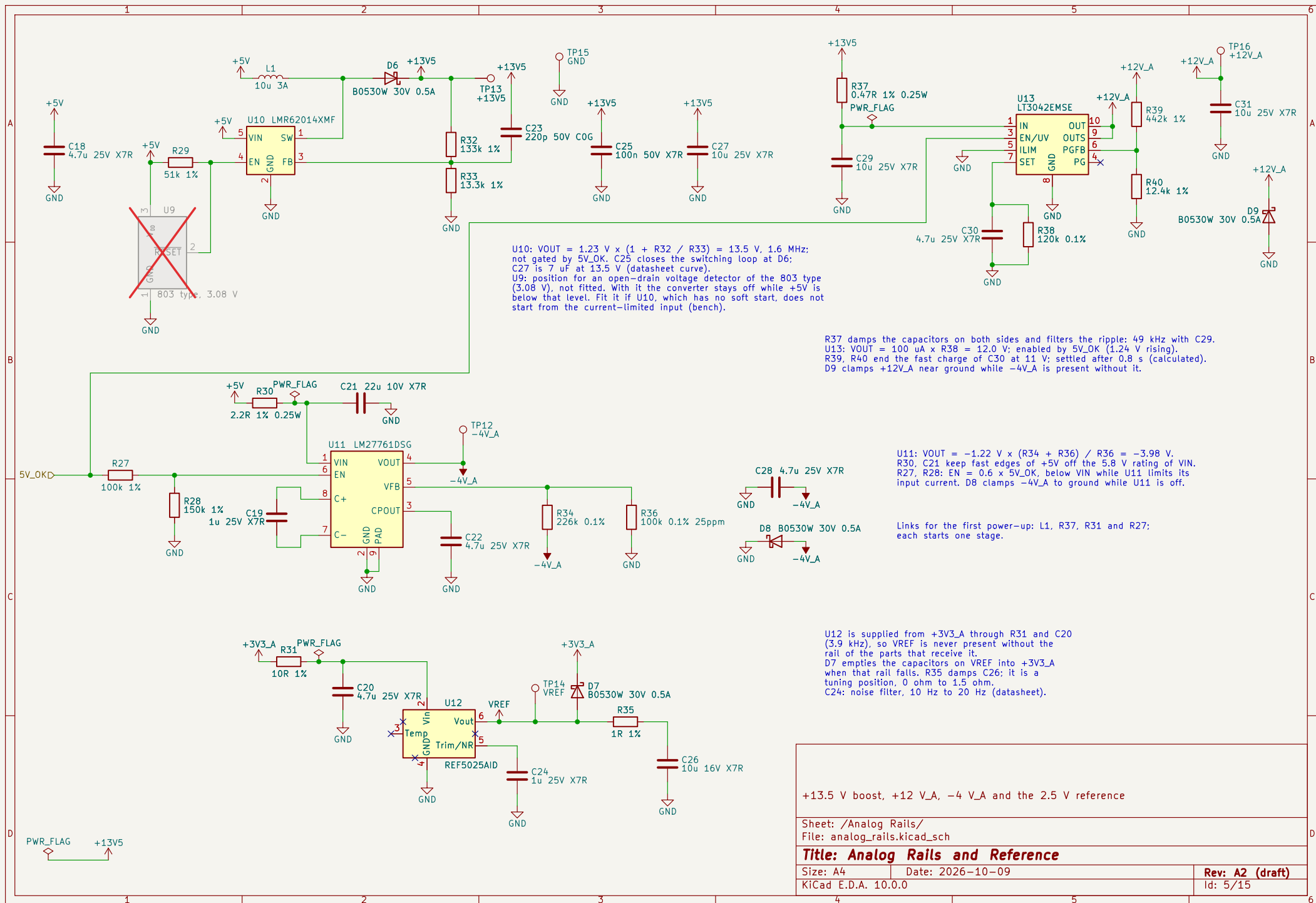
U7: +3V3_C, logic of the carrier. U8: +3V3_A, ADC, comparators, DAC, monitor converter. The regulators are specified for 10 uF at most on the output (datasheet). Over all sheets +3V3_A holds 3.8 uF and 4.7 uF more behind the 10 ohm of the reference supply. +3V3_C 2.4 uF (calculated). A disabled regulator empties its output through 230 ohm (datasheet), so +3V3_A starts from 0 V after every dip of the rail. R26 and D5: power indicator, 1.3 mA. R23 bleeds +5V when no source is connected. H1 to H4: mounting holes, tied to ground. FID1 to FID3: fiducials for the placement machine.

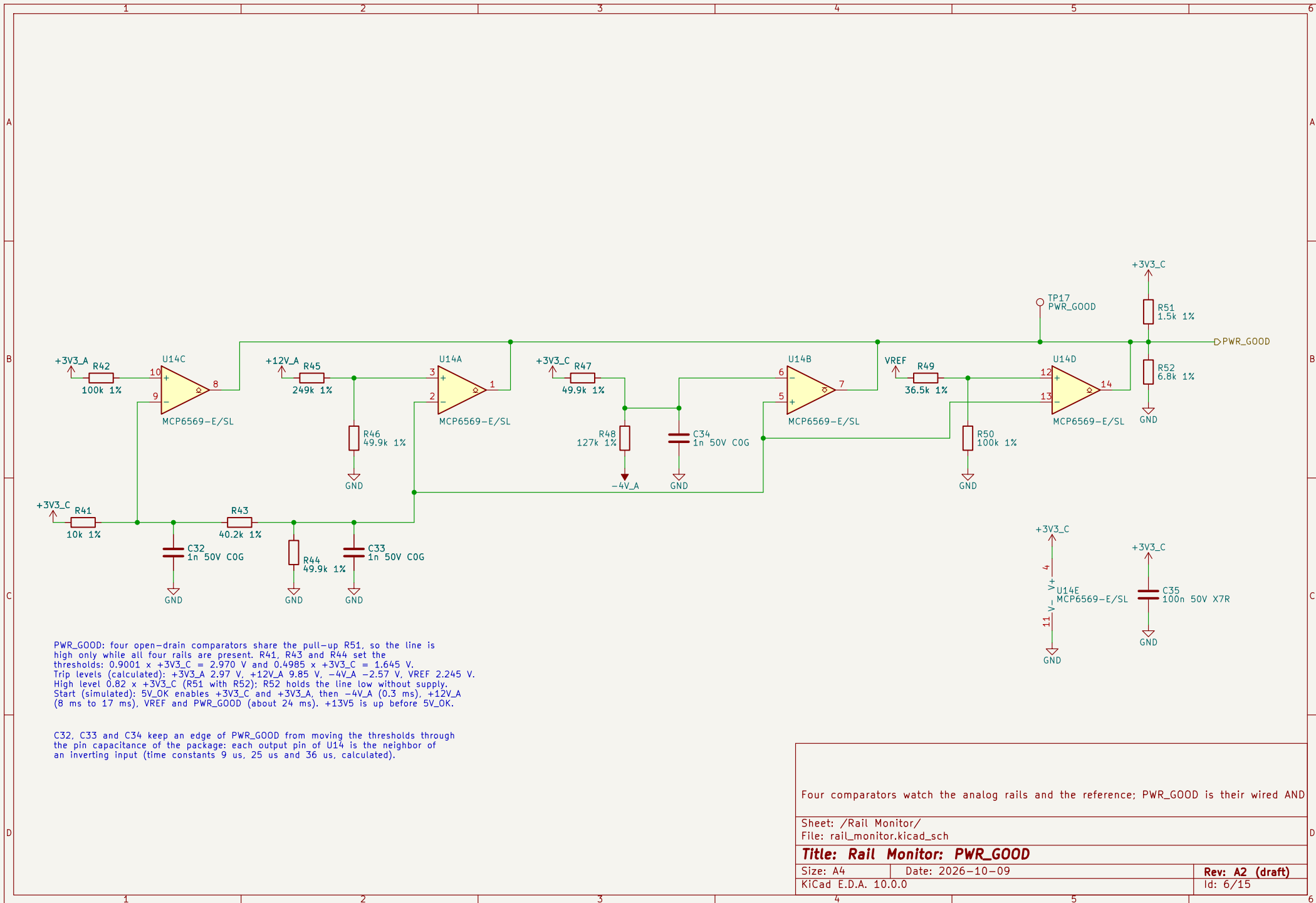
Supervisor of the 5 V rail; its output 5V_OK enables the two 3.3 V regulators

Sheet: /Logic Supplies/
File: logic_supplies.kicad_sch

Title: Logic Supplies and Rail Supervisor

Size: A4 Date: 2026-10-09 Rev: A2 (draft)
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Enable: SMU_ON reaches U16 through Q1 only while 5V_OK is high. When +5V falls below 3.83 V to 4.00 V the supervisor opens Q1 and R53 pulls the pin low: the pre-regulator is shed without the firmware. With +5V at 4.25 V the pin gets 2.36 V (calculated, gate threshold estimated; it needs 1.2 V, datasheet). Start: DAC at the 0.80 V code for 200 ms with SMU_ON low, then SMU_ON high; the output switch closes last.

U16 land pattern: lead pads agree with TI drawing 4210895-2/E; center pad 1.7 x 3.3 mm against 1.58 x 2.85 mm accepted; via array in the pad at layout.

Set-point: U15 runs at gain 2 from $V_{REF} / 2$ (R55, R56), reference input buffered: frame 0x5000 | code. U17 has a gain of 2.1: $LDO_OUT = 10 \text{ mV} + \text{code} \times 1.2817 \text{ mV}$ before calibration, and no frame gives more than 5.26 V (5.39 V at the tolerance limits, calculated). R54 and C41: 10 ms, so the set-point moves at 0.52 V/ms at most.

Pre-regulator: U16 holds PRE_FB at 0.5 V and runs in forced PWM (PS/SYNC high). U19 makes $PRE_FB = 0.200 \times V_{PRE} + 0.146 \times V_{REF} - 0.191 \times LDO_OUT$, so $V_{PRE} = 0.672 \text{ V} + 0.956 \times LDO_OUT$ (calculated): 0.64 V above the output at 0.8 V and 0.45 V at 5.0 V, also while a device holds the output above the set-point. A short at the output folds V_{PRE} back to 0.84 V (simulated). R65, R64 and C55: 0.62 ms low-pass that keeps the sense out of the loop of U16. D13 lets V_{PRE} follow a fast rise of the output at once, C52: phase lead for U19. R62 empties C42, C45 and C50 within about 0.2 s while U16 is off (calculated).

Linear regulator: U18 takes IN from V_{PRE} through FB1, VCONTROL from +13V5 through R61 and D10, and SET from U17. Both supply pins are rated 0.3 V below OUT (datasheet); D11 carries the current while OUT is above IN, and D10 lets C47 hold VCONTROL up at power-off. D12 keeps LDO_OUT above about -0.2 V (estimate). R69 is the minimum load: 3.7 mA at 0.8 V, 6.9 mA at 5.0 V (calculated); it also takes the leakage of D11. C43 and R59 damp the resonance of FB1 with C46. Design current: 1.0 A up to 2.0 V, falling to 0.6 A at 5.0 V (calculated, not yet confirmed on a board). TP25, TP20: probe grounds at the linear regulator and at the pre-regulator.

On the board: R67 and R62 connect at pin 1 of FB1, R65 and D13 at the OUT pins of U18; R55 sits next to R56 and R57 next to R58.

Pre-regulator that tracks the output, linear regulator with clamps, 12-bit set-point

Sheet: /Source Meter/
File: source_meter.kicad_sch

Title: Source Meter: Pre-regulator, LDO and DAC

Size: A4 Date: 2026-10-09

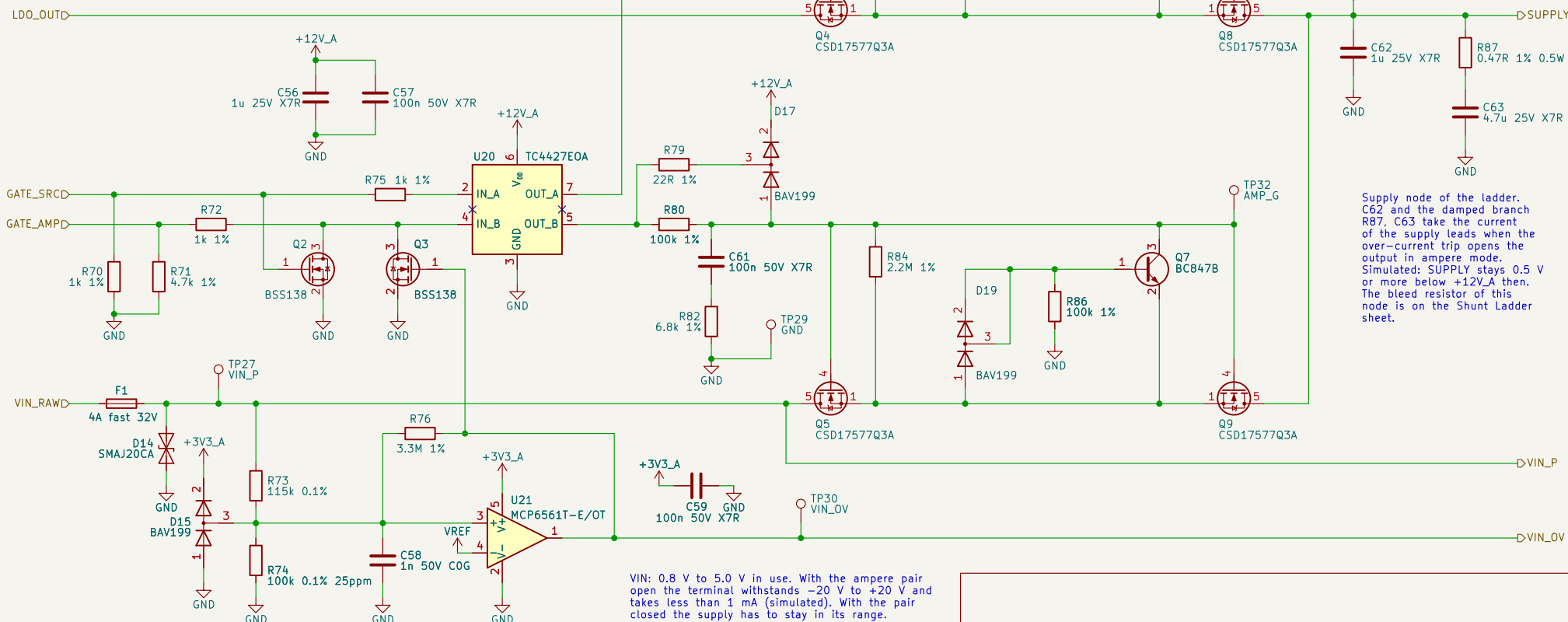
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Mode pairs: Q4, Q8 pass the regulator output (source mode), Q5, Q9 pass VIN (ampere mode). Both are open at reset. The gate networks are alike: for the ampere pair: Closing: R80 charges the gate against C61, R82: $(100\text{ k} + 6.8\text{ k}) \times 100\text{ nF} = 10.7\text{ ms}$, after a first step of 0.76 V, below the threshold (calculated). The pair closes as a source follower: SUPPLY rises with about 0.9 V/ms, full gate drive in 40 ms (simulated). Opening: D17 (pin 1 to 3) and R79 discharge the gate at once: one pair is open before the other conducts. D17 (pin 3 to 2) bounds the gate at +12V_A plus two drops. Q7 joins gate and common source while that node is below ground: an open pair stays open with a reversed supply. D19 (pins 2 and 3 joined) holds the base one diode drop below the emitter otherwise. R84 returns the gate to the common source and keeps 95.7 % of the drive (calculated); no resistor goes from a gate to ground. TP32 and TP31 sit on 100 k nodes: a x10 probe lowers the gate by about 1 % (calculated). TP28, TP29: probe grounds at the two mode pairs.

Requests: R70 and R71 hold GATE_SRC and GATE_AMP low while the controller pins float. Q2 holds the ampere request low at U20 while source mode is requested, Q3 while VIN_OV is high: the two pairs are never closed together, whatever the controller asks. R75 and R72 limit the current into U20 while +12V_A is absent (2.7 mA, calculated). Firmware leaves 5 ms between lowering one request and raising either; a pair counts as closed 40 ms after its request, and the output switch closes after the mode pair.



Supply node of the ladder. C62 and the damped branch R87, C63 take the current of the supply leads when the over-current trip opens the output in ampere mode. Simulated: SUPPLY stays 0.5 V or more below +12V_A then. The bleed resistor of this node is on the Shunt Ladder sheet.

VIN: 0.8 V to 5.0 V in use. With the ampere pair open the terminal withstands -20 V to +20 V and takes less than 1 mA (simulated). With the pair closed the supply has to stay in its range. D14 takes hot-plug rings and the kick of the supply leads; it conducts from 22.2 V (datasheet): never a 24 V supply. F1 does not protect D14. Detector: VIN_OV goes high at 5.46 V and low again at 5.35 V (calculated; 2.5 V at the tap of R73 and R74), 5.41 V to 5.51 V and 5.30 V to 5.40 V with the tolerances. Not latched. D15 keeps the input within a diode drop of both rails: R73 limits its current to 0.17 mA at 20 V (calculated).

Source and ampere mode switches with their gate networks, VIN input and its protection

Sheet: /Path Switching/

File: path_switching.kicad_sch

Title: Mode Switches and VIN Protection

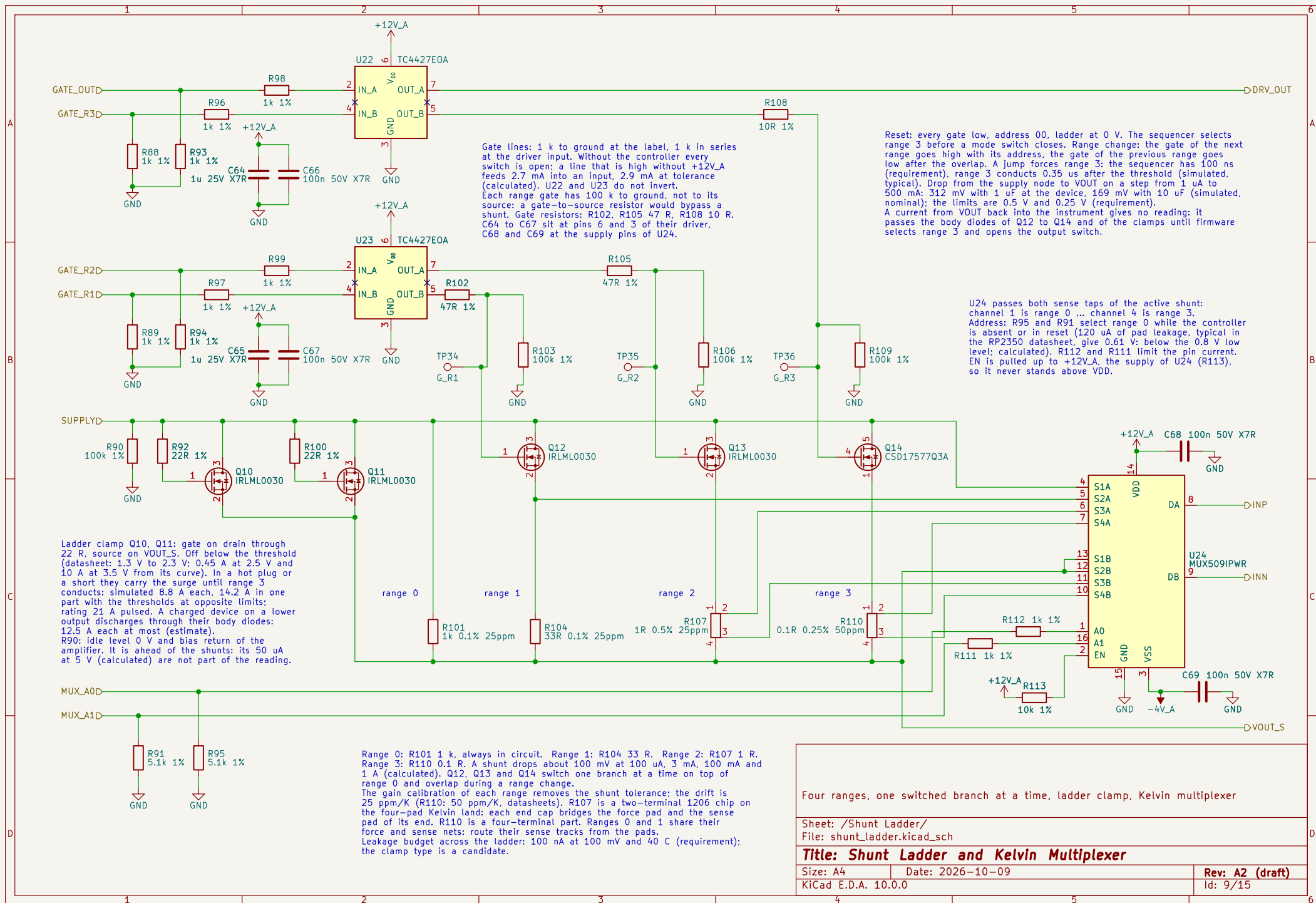
Size: A4

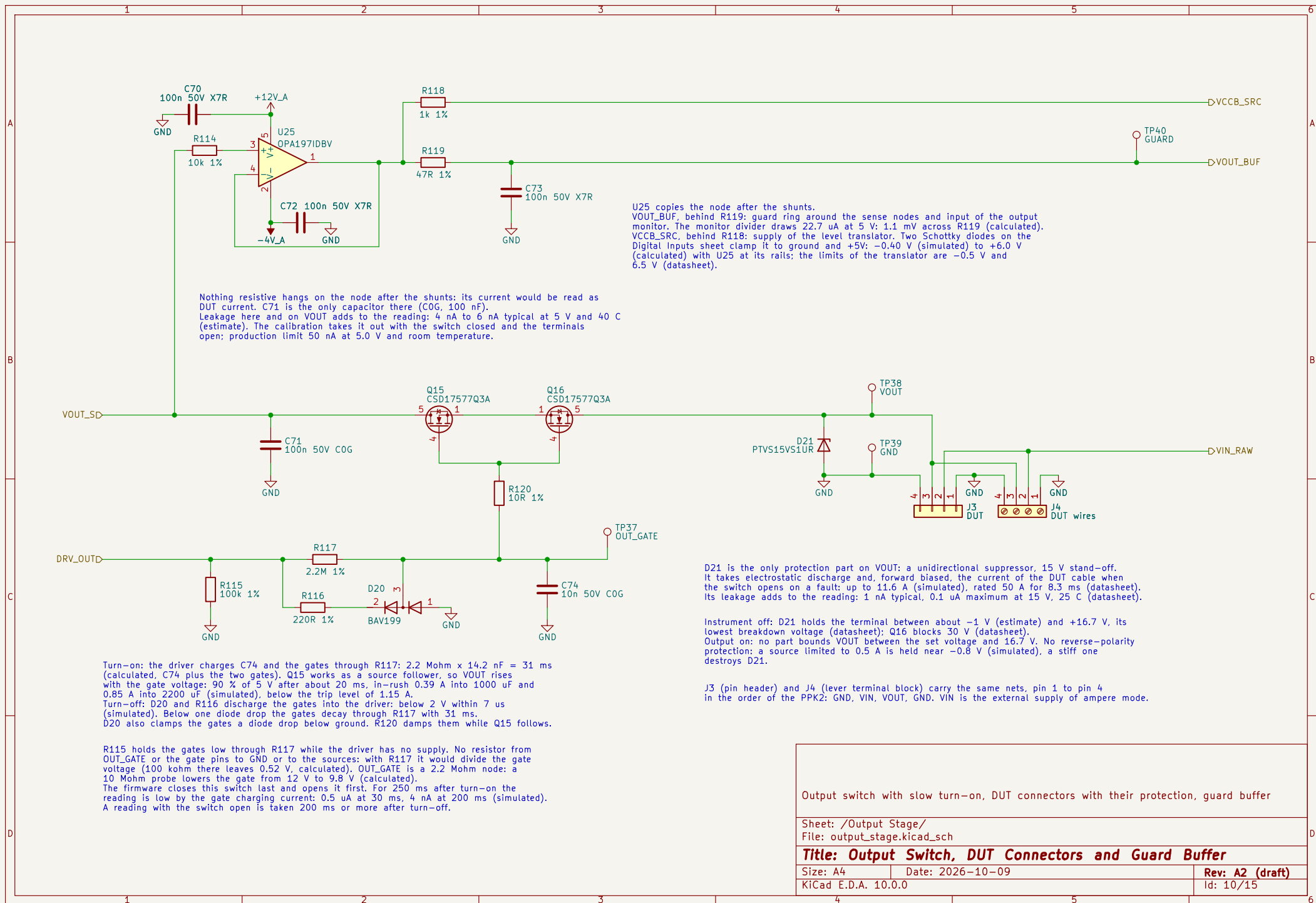
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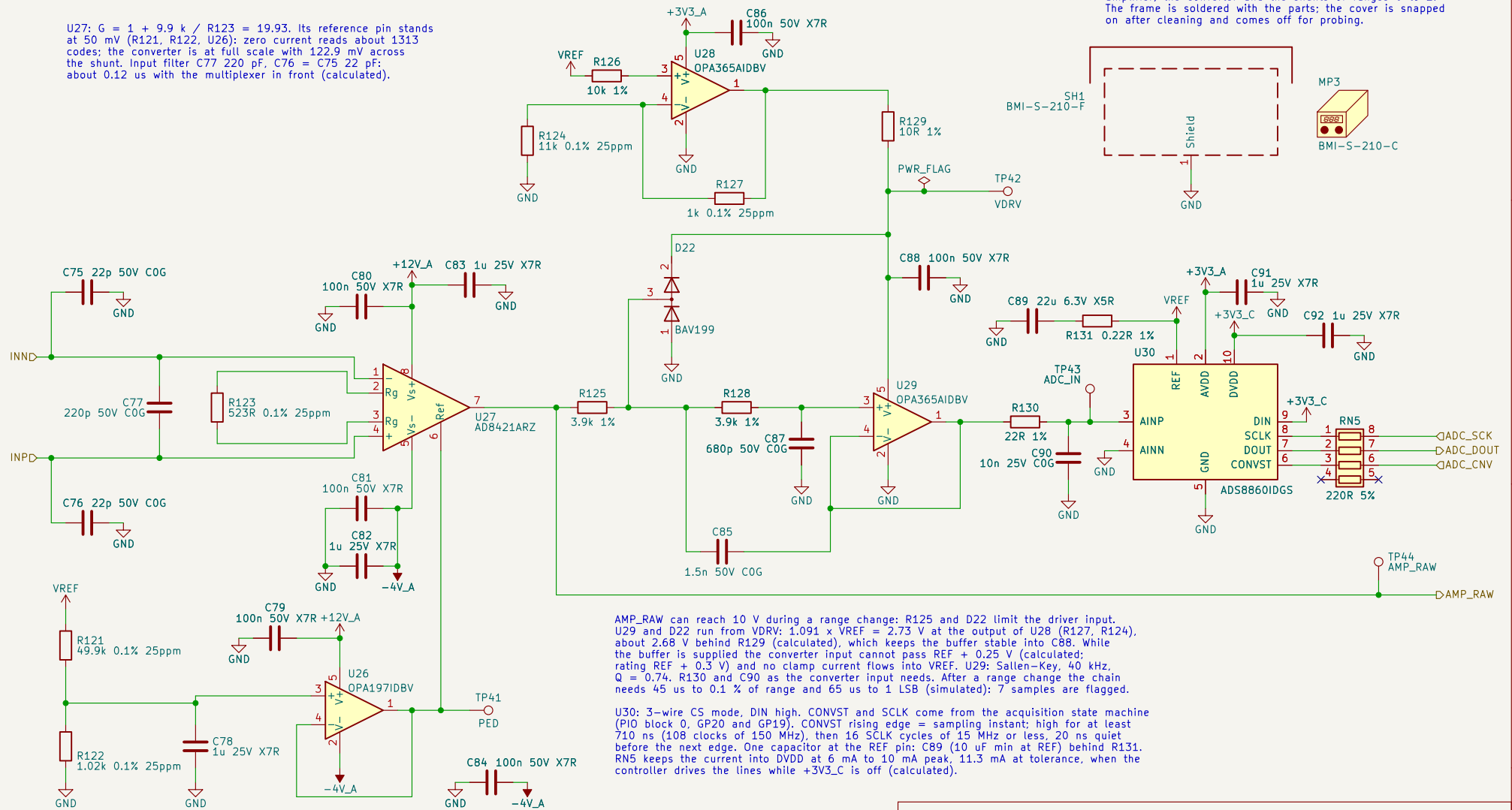
Output switch with slow turn-on, DUT connectors with their protection, guard buffer

Sheet: /Output Stage/
File: output_stage.kicad_sch

Title: Output Switch, DUT Connectors and Guard Buffer

Size: A4	Date: 2026-10-09	Rev: A2 (draft)
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U27: $G = 1 + 9.9 \text{ k} / R_{123} = 19.93$. Its reference pin stands at 50 mV (R121, R122, U26): zero current reads about 1313 codes; the converter is at full scale with 122.9 mV across the shunt. Input filter C77 220 pF, C76 = C75 22 pF: about 0.12 μ s with the multiplexer in front (calculated).



Gain 19.93, 50 mV pedestal, limiter on VDRV, 40 kHz two-pole filter, 16-bit converter

Sheet: /Signal Chain/
File: signal_chain.kicad_sch

Title: Amplifier, Anti-alias Filter and ADC

Size: A4
KiCad E.D.A. 10.0.0

Date: 2026-10-09

Rev: A2 (draft)
Id: 11/15

